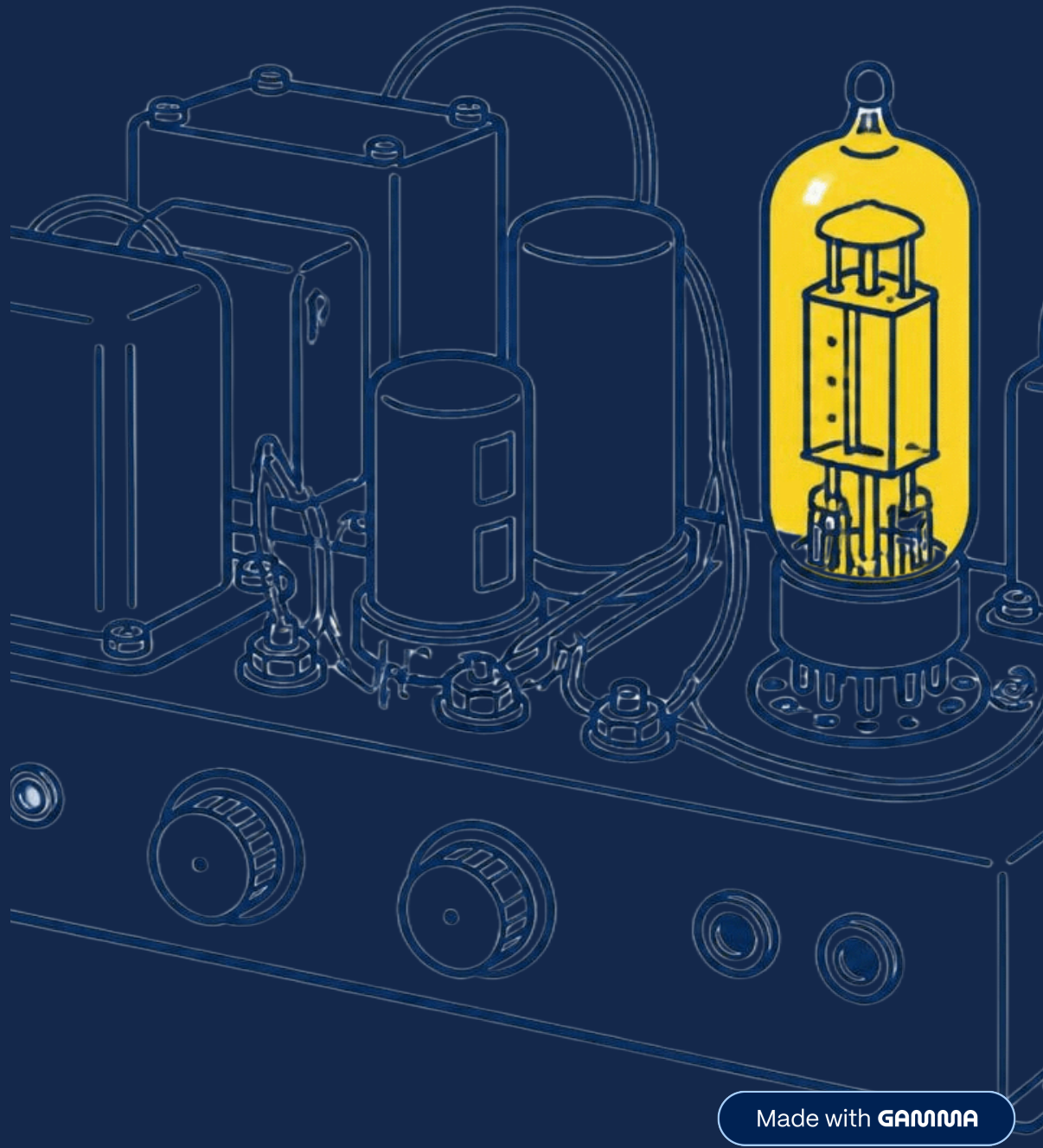
Section 1 · 1950 Proposal

Can machines think?

Turing replaces contested definitions with an operational question of intellectual performance.

A. M. Turing · Computing Machinery and Intelligence, Mind (1950)



Made with GAMMA

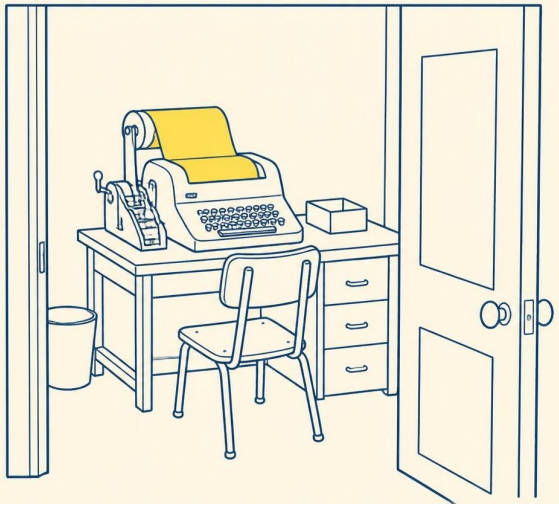
The imitation game changes the question

Section 1 · Replacing the man with a machine

The original party game

An interrogator questions a man and a woman in separate rooms to identify their sexes correctly.

Baseline setup



Man replaced by a digital computer

The mechanical substitution

A digital computer replaces the man. Does the interrogator err as often when a machine plays?

Turing's new question



Made with GAMMA



Conversation isolates intellectual performance

Section 2 · Critique of the new problem

A separated interrogator exchanges typed questions and answers with a human and a machine.

Without appearance, voice tone or biological identity, the test evaluates conversational mental capacity without requiring human physical embodiment.

Pure symbolic inquiry

Made with GAMMA

One computer can imitate many machines

Section 4 · Digital computers and universality

Store

Information capacity holding instructions, tables, and working intermediate variables.

→

Executive unit

The central mechanism carrying out individual elementary operations in sequence.

→

Control

The system ensuring operating instructions execute strictly according to programmed order.

Universal imitation

Made with GAMMA

Turing predicted progress, not proof

70%

maximum expected accuracy for an average interrogator after five minutes of questioning

~50 years

conjectured timeframe from 1950 to reach this benchmark

---

109 bits

projected storage capacity required for the machine

Conjecture

Section 6: a stated belief about future performance, not a mathematical certainty

Made with GAMMA

Mathematical limits do not settle the debate

Section 6(3) · The Mathematical Objection

Provable machine limitations

Gödel's theorem and Turing's halting results show discrete-state systems cannot resolve every formal statement.

Formal limits exist

Unproved human superiority

Turing observes it has not been proved that human intellect is free from comparable internal limitations.

Turing's counterpoint

Made with GAMMA

Consciousness remains an open mystery

> **"The only way to know that a machine thinks is to be the machine and to feel oneself thinking."**

Professor Jefferson's Lister Oration (1949), quoted by Turing in Section 6(4)

Solipsist impasse

Turing notes demanding subjective experience leads to solipsism: we credit other humans with thought by communication, not direct introspection.

Section 6(4)

Made with GAMMA

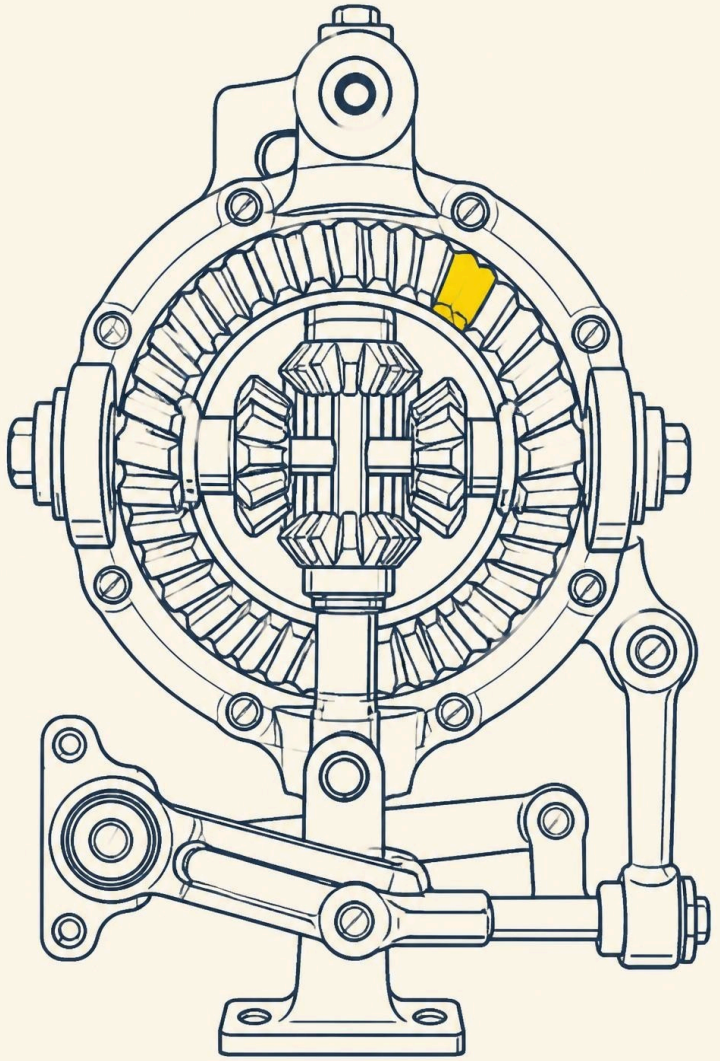
Machines may surprise their makers

Section 6(6) · Lady Lovelace’s objection

Lady Lovelace claimed the Analytical Engine had no pretensions to originate anything, executing only programmed commands.

Turing replies that machines continually surprise their operators by yielding consequences not consciously foreseen.

Unexpected behavior



Made with GAMMA

Teach a child machine rather than program an adult

Section 7 · Learning machines and developmental architecture

1 Child programme

Construct a simple initial mechanism with minimal fixed knowledge and high plasticity.

2 Education and trial

Subject the machine to experience, rewards, punishments, and systematic feedback.

3 Altered rules

Learned rules change internal behavior while underlying machine constitution remains fixed.

Evolutionary learning

Made with GAMMA

Section 7 · Conclusion

An experiment and an open agenda

Turing proposed empirical investigation rather than dogmatic proof, urging work on chess and language learning.

"We can only see a short distance ahead, but we can see plenty there that needs to be done."



Made with GAMMA